

## CS 2490 – Homework 3

Turn in by 2025/09/12 11:59PM

You **must** turn in your assignment as a single zip file entitled: *lastname\_firstname\_hw3.zip*. It will contain a pdf file *lastname\_firstname\_hw3.pdf* with your written answers (for which you can use L<sup>A</sup>T<sub>E</sub>X, etc. or you can write your solutions by hand and scan them), and a JSON file for each relevant exercise named *lastname\_firstname\_hw3\_exX.json* (where X is the exercise number).

**Exercise 1.** The alphabet in this exercise is  $\Sigma = \{a, b\}$ . Draw a NFA that recognizes the language of all words that contain the substring *aba anywhere* in the word.

Use the tool online to test your answer and export the JSON file that you will include in your submission.

**Exercise 2.** Draw the *determinization* of the NFA you drew in the previous exercise.<sup>1</sup>

Use the tool online (the one for DFA of course!) to test your answer and export the JSON file that you will include in your submission.

<sup>1</sup> Consistency will be taken into account. Thus, even if the previous exercise is not correct, you still get all credit for this exercise if you correctly determinize your NFA.

**Exercise 3.** The alphabet in this exercise is  $\Sigma = \{0, 1\}$ . Draw a NFA that recognizes the language of all words in which the substring 101 appears at least twice (possibly overlapping).

Use the tool online to test your answer and export the JSON file that you will include in your submission.

**Exercise 4.** Let  $\Sigma$  be any alphabet. Let  $w$  be a word over  $\Sigma$ . Define a NFA  $\mathcal{A}_w$  over  $\Sigma$  that accepts  $w$  and only  $w$ . In other words,  $\mathcal{A}_w$  recognizes the language  $\{w\}$ .

**Exercise 5.** Let  $\Sigma$  be any alphabet. Let  $L$  be a finite language over  $\Sigma$ , that is,  $L$  contains only a finite number of words. Using the previous exercise,<sup>2</sup> prove that there is a NFA over  $\Sigma$  that recognizes the language  $L$ .

<sup>2</sup> If you didn't complete the previous exercise, you can still use the result it states.